

# EC2 Creation and Linux

## Class no: 8

### Elastic Compute Cloud (EC2):

Amazon EC2 (Elastic Compute Cloud) is a core AWS service that provides on-demand, resizable virtual servers in the cloud. You can launch as many or as few instances as needed, configure CPU, memory, storage, and networking, and scale up or down based on demand.

### **Key Features of Amazon EC2**

- Instances – Virtual servers for running applications or databases.
- Amazon Machine Images (AMIs) – Preconfigured OS and software templates.
- Instance Types – Different CPU, memory, and storage configurations.
- Amazon EBS Volumes – Persistent block storage for data.
- Security Groups – Virtual firewalls controlling inbound/outbound traffic.
- Elastic Load Balancing & Auto Scaling – Distribute traffic and adjust capacity automatically.

### Hands On: How to create an instance in EC2 and make connection.

#### **I.Windows OS**

**Step 1:** Go to the AWS account, search EC2, Click launch instance.

- Instance name: windowsinstanceforpractice
- Type: Windows
- Size: t3micro

Launch an instance

Name  
windowsinstanceforpractice [Add additional tags](#)

▼ **Application and OS Images (Amazon Machine Image)** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Q Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

aws Mac ubuntu Microsoft Red Hat SUSE debian

[Browse more AMIs](#)  
Including AMIs from AWS, Marketplace and the Community

**Amazon Machine Image (AMI)**

Microsoft Windows Server 2025 Base Free tier eligible

ami-0f23ee11d840e3e64 (64-bit (x86))  
Virtualization: hvm ENA enabled: true Root device type: ebs

**Description**

Microsoft Windows 2025 Datacenter edition, [English]

Microsoft Windows Server 2025 Full Locale English AMI provided by Amazon

| Architecture | AMI ID                | Publish Date | Username      |
|--------------|-----------------------|--------------|---------------|
| 64-bit (x86) | ami-0f23ee11d840e3e64 | 2026-03-12   | Administrator |

[Verified provider](#)

▼ **Instance type** [Info](#) | [Get advice](#)

Instance type

t3.micro Free tier eligible

Family: t3 2 vCPUs 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0108 USD per Hour  
On-Demand Windows base pricing: 0.02 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour  
On-Demand Intel base pricing: 0.0396 USD per Hour On-Demand SUSE base pricing: 0.0108 USD per Hour

☒ All generations [Compare instance types](#)

- Create key pair -> provide a name: batchfrw ->download the file.

## ▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

### Key pair name - required

batchfrw

[Create new key pair](#)

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

- Choose launch instance.

## ▼ Summary

Number of instances [Info](#)

1

### Software Image (AMI)

Microsoft Windows Server 2025 ...[read more](#)  
ami-0f23ee11d840e3e64

### Virtual server type (instance type)

t3.micro

### Firewall (security group)

New security group

### Storage (volumes)

1 volume(s) - 30 GiB

[Cancel](#)

[Launch instance](#)

[Preview code](#)

**Step 2:** Now the instance will be visible on the console. Choose the windows instance, click connect.

## Instances (1/2) [Info](#)

Last updated  
less than a minute ago

[Connect](#)

[Instance state](#) ▼

[Actions](#) ▼

[Launch instances](#)

All states ▼

< 1 >



| <input type="checkbox"/>            | Name <a href="#">✎</a> | Instance ID         | Instance state <a href="#">▼</a>              | Instance type <a href="#">▼</a> | Status check      | Alarm status                  | Availability Zone |
|-------------------------------------|------------------------|---------------------|-----------------------------------------------|---------------------------------|-------------------|-------------------------------|-------------------|
| <input type="checkbox"/>            | linuxmachinef...       | i-0b954ba2ea9cc11f4 | ✓ Running <a href="#">ⓘ</a> <a href="#">⌵</a> | t3.micro                        | ⌚ Initializing    | <a href="#">View alarms +</a> | eu-north-1b       |
| <input checked="" type="checkbox"/> | windowsinsta...        | i-0139413f8eddf5f1  | ✓ Running <a href="#">ⓘ</a> <a href="#">⌵</a> | t3.micro                        | ✓ 3/3 checks pass | <a href="#">View alarms +</a> | eu-north-1b       |

**Step 3:** Move to RDP client Tab.

[Instances](#) > [i-0139413f8eddf5f1](#) > Connect to instance

## Connect [Info](#)

Connect to an instance using the browser-based client.

Instance ID  
[i-0139413f8eddf5f1](#) (windowsinstanceforpractice)

VPC ID  
[vpc-08d23a3e5bc324ead](#)

Security groups  
[sg-0fed0bd95fa25abb5](#) (launch-wizard-2)

IAM role  
-

SSM Session Manager [RDP client](#) EC2 serial console

[Record RDP connections](#)

You can now record RDP connections using AWS Systems Manager just-in-time node access. [Learn more](#)

Instance ID  
[i-0139413f8eddf5f1](#) (windowsinstanceforpractice)

### Connection Type

[Connect using RDP client](#)  
Download a file to use with your RDP client and retrieve your password.

[Connect using Fleet Manager](#)  
To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be installed and running. For more information, see [Working with SSM Agent](#).

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:

Public DNS  
[ec2-16-170-245-6.eu-north-1.compute.amazonaws.com](#)

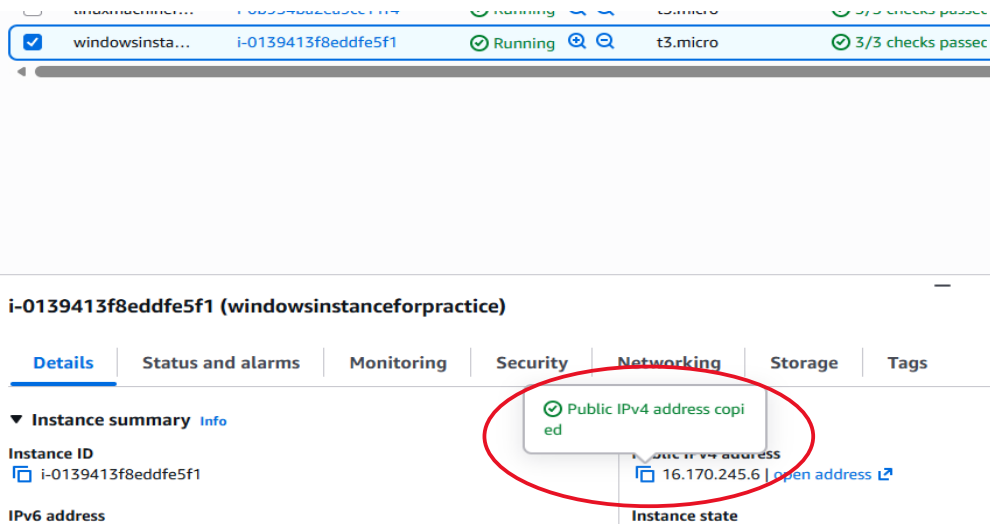
Username [Info](#)

[Administrator](#)

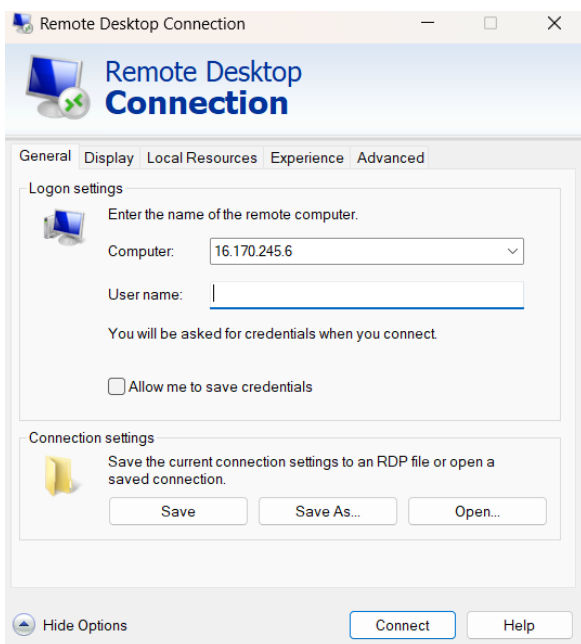
Password [Get password](#)

[If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.](#)

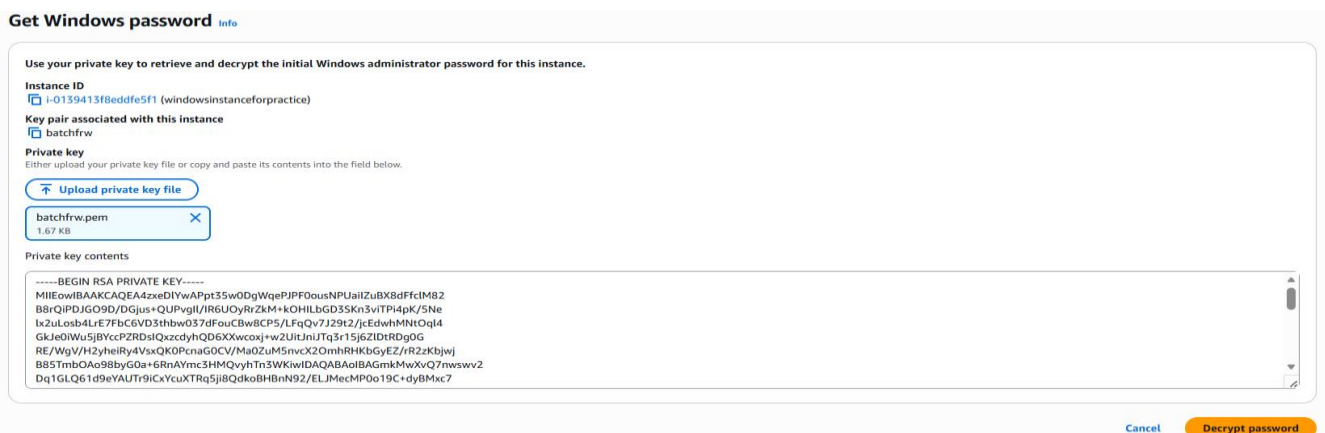
**Step 4:** Copy the public Ip address of our windows EC2 instance. Now in your laptop search RDC (Remote Desktop Connection).



- Paste the Ip address in computer field.



- Now we will get the decrypted password. In the RDP page of the Windows instance, scroll down. There is an option to **get password**. Click on it. Upload our downloaded **key pair file (batchfrw)**. Choose **Decrypt Password**.



- Now we can see the respective username and password on the RDP page.

SSM Session Manager

RDP client

EC2 serial console

Record RDP connections

You can now record RDP connections using AWS Systems Manager just-in-time node access. [Learn more](#)

Instance ID

i-0139413f8eddfef1 (windowsinstanceforpractice)

Connection Type

Connect using RDP client

Download a file to use with your RDP client and retrieve your password.

Connect using Fleet Manager

To connect to the instance using Fleet Manager Remote Desktop Client, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

Download remote desktop file

When prompted, connect to your instance using the following username and password:

Public DNS

ec2-16-170-245-6.eu-north-1.compute.amazonaws.com

Username Info

Administrator

Password

V6Er1hYv2lbuH%Y.55G4qPhbCLJIRxHV

If you've joined your instance to a directory, you can use your directory credentials to connect to your instance.

- Enter it and click connect.

**Note:** We can also connect by downloading the remote desktop file. Once downloaded it will ask for password as the username will be auto populated. Enter it and click connect.

Remote Desktop Connection

Remote Desktop Connection

General Display Local Resources Experience Advanced

Logon settings

Enter the name of the remote computer.

Computer: 16.170.245.6

User name: Administrator

You will be asked for credentials when you connect

☒ Allow me to save credentials

Connection settings

Save the current connection settings to an RDP file or open a saved connection.

Save Save As... Open...

Hide Options

Connect

Help

Windows Security

Enter your credentials

These credentials will be used to connect to 16.1

Administrator

Password

DESKTOP-NBCS2S9\Administrator

☒ Remember me

More choices

OK

A windows virtual machine using remote connection will be created.



## II. Linux OS

**Step 1:** Create a new ec2 instance with the name: linuxmachineforpractice, choose OS type as AWS Linux and use the existing key pair.

### Name and tags [Info](#)

Name  [Add additional tags](#)

### ▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

[Recents](#) | [Quick Start](#)



#### Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI  
ami-0aaa636894689fa47 (64-bit (x86), uefi-preferred) / ami-0ecd046e71cbe511e (64-bit (Arm), uefi)  
Virtualization: hvm ENA enabled: true Root device type: ebs

#### Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260302.1 x86\_64 HVM kernel-6.1

| Architecture | Boot mode      | AMI ID                | Publish Date | Username |                                   |
|--------------|----------------|-----------------------|--------------|----------|-----------------------------------|
| 64-bit (x86) | uefi-preferred | ami-0aaa636894689fa47 | 2026-03-03   | ec2-user | <a href="#">Verified provider</a> |

### ▼ Instance type [Info](#) | [Get advice](#)

#### Instance type

t3.micro  
Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0108 USD per Hour  
On-Demand Windows base pricing: 0.02 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour  
On-Demand RHEL base pricing: 0.0396 USD per Hour On-Demand SUSE base pricing: 0.0108 USD per Hour

[Additional costs apply for AMIs with pre-installed software](#)

☐ All generations

[Compare instance types](#)

### ▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*  [Create new key pair](#)

### ▼ Summary

Number of instances [Info](#)

1

#### Software Image (AMI)

Amazon Linux 2023 AMI 2023.10...[read more](#)  
ami-0aaa636894689fa47

Virtual server type (instance type)  
t3.micro

Firewall (security group)  
New security group

Storage (volumes)  
1 volume(s) - 8 GiB

[Cancel](#)

[Launch instance](#)

[Preview code](#)

**Step 2:** Now select our Linux instance and click connect.

Instances (1/2) Info

Find Instance by attribute or tag (case-sensitive)

All states ▾

Last updated 34 minutes ago

Refresh

Connect

| <input type="checkbox"/>            | Name             | Instance ID         | Instance state | Instance type | Status check      | Alarm status                  | Availability Zone |
|-------------------------------------|------------------|---------------------|----------------|---------------|-------------------|-------------------------------|-------------------|
| <input checked="" type="checkbox"/> | linuxmachinef... | i-0b954ba2ea9cc11f4 | Running        | t3.micro      | 3/3 checks passed | <a href="#">View alarms +</a> | eu-north-1b       |
| <input type="checkbox"/>            | windowsinst...   | i-0139413f8eddf5f1  | Running        | t3.micro      | 3/3 checks passed | <a href="#">View alarms +</a> | eu-north-1b       |

**Step 3:** Now it provides an instance connect page. Click connect.

**Connect** [Info](#)

Connect to an instance using the browser-based client.

**Instance ID**  
 i-0b954ba2ea9cc11f4 (linuxmachineforpractice)

**VPC ID**  
 vpc-08d23a3e5bc324ead

**Security groups**  
 sg-07baf94d0e9da4454 (launch-wizard-3)

**IAM role**  
-

**EC2 Instance Connect**

SSM Session Manager

SSH client

EC2 serial console

**Instance ID**  
 i-0b954ba2ea9cc11f4 (linuxmachineforpractice)

**Connection type**

☒ **Connect using a Public IP**  
Connect using a public IPv4 or IPv6 address

☐ **Connect using a Private IP**  
Connect using a private IP address and a VPC endpoint

☒ **Public IPv4 address**  
 51.20.4.78

☐ **IPv6 address**

**Username**  
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ec2-user.

 **Note:** In most cases, the default username, ec2-user, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel

Connect

**Step 4:** A Linux virtual machine will be created.

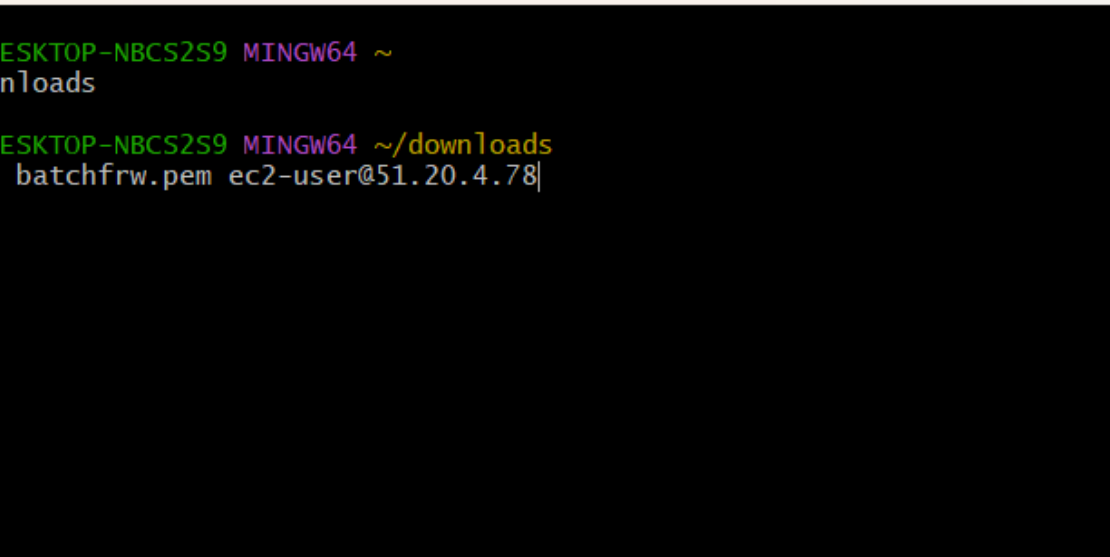
The screenshot shows a web browser window with multiple tabs open. The active tab displays the AWS Management Console terminal for an EC2 instance named "eu-north-1". The terminal output shows the Amazon Linux 2023 logo, the URL "https://aws.amazon.com/linux/amazon-linux-2023", and a series of commands executed by the user "ec2-user":  

```
[ec2-user@ip-172-31-46-238 ~]$ ls  
[ec2-user@ip-172-31-46-238 ~]$ mkdir newuser  
[ec2-user@ip-172-31-46-238 ~]$ ls  
newuser  
[ec2-user@ip-172-31-46-238 ~]$ cd .  
[ec2-user@ip-172-31-46-238 ~]$ cd  
[ec2-user@ip-172-31-46-238 ~]$ pwd  
/home/ec2-user  
[ec2-user@ip-172-31-46-238 ~]$
```

The browser's address bar shows the URL: <https://eu-north-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0b954ba2ea9cc11f4&osUser=ec2-user&region...>

Below the terminal, the instance ID **i-0b954ba2ea9cc11f4 (linuxmachineforpractice)** is displayed, along with its Public IP (51.20.4.78) and Private IP (172.31.46.238).

For example:

- 
- The screenshot shows a Windows terminal window titled "MINGW64:/c:/Users/nithya/downloads". The terminal output is as follows:
- ```
nithya@DESKTOP-NBCS2S9 MINGW64 ~  
$ cd downloads  
  
nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads  
$ ssh -i batchfrw.pem ec2-user@51.20.4.78|
```
- The terminal window has a black background with green and white text. The prompt "nithya@DESKTOP-NBCS2S9" is green, and "MINGW64" is white. The tilde "~" represents the home directory. The command "cd downloads" is entered, and the prompt changes to "nithya@DESKTOP-NBCS2S9 MINGW64 ~/downloads". The command "ssh -i batchfrw.pem ec2-user@51.20.4.78|" is entered, and the cursor is at the end of the line.

- Note:** If the connection is made via putty ensure that we are creating .ppk key pair file while generating it. Because putty does not support .pem key pair file.

```
ec2-user@ip-172-31-46-238:~  
nithya@DESKTOP-NBCS259 MINGW64 ~  
$ cd downloads  
  
nithya@DESKTOP-NBCS259 MINGW64 ~/downloads  
$ ssh -i batchfrw.pem ec2-user@51.20.4.78  
The authenticity of host '51.20.4.78 (51.20.4.78)' can't be established.  
ED25519 key fingerprint is: SHA256:aiBk9YBN8IgeiNGjboxLRJ74EGoeDF9+HgPIpTgSj64  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? y  
Please type 'yes', 'no' or the fingerprint: yes  
Warning: Permanently added '51.20.4.78' (ED25519) to the list of known hosts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
  
#_      Amazon Linux 2023  
~\#####  
~\#####  
~\###|  
~\#/V-> https://aws.amazon.com/linux/amazon-linux-2023  
~~~~~  
~-./-/-/  
_-/_/_/m/'-/-/
```

Last login: Wed Mar 18 13:55:10 2026 from 13.48.4.203  
[ec2-user@ip-172-31-46-238 ~]\$ |